ALGEBRA 2 · TRIGONOMETRIC FUNCTIONS

Unit Circle to Sine Wave

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Unit Circle to Sine Wave** on Mr. K's Classroom: Algebra 2 → Trigonometric Functions → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. At 90°, what is the height of the point on the unit circle?

   1

2. Is sin 210° positive or negative?

   Negative: the point is below the x-axis.

2 Explore use the tool and record what you see

Unit circle values

Drag the point to each angle.

θ (degrees)	θ (radians)	cos θ	sin θ	Quadrant
0°	**0**	**1**	**0**	**on an axis**
30°	**$\frac{\pi}{6}$**	**$\frac{\sqrt{3}}{2}$**	**$\frac{1}{2}$**	**I**
45°	**$\frac{\pi}{4}$**	**$\frac{\sqrt{2}}{2}$**	**$\frac{\sqrt{2}}{2}$**	**I**
90°	**$\frac{\pi}{2}$**	**0**	**1**	**on an axis**
150°	**$\frac{5\pi}{6}$**	**$-\frac{\sqrt{3}}{2}$**	**$\frac{1}{2}$**	**II**
225°	**$\frac{5\pi}{4}$**	**$-\frac{\sqrt{2}}{2}$**	**$-\frac{\sqrt{2}}{2}$**	**III**
300°	**$\frac{5\pi}{3}$**	**$\frac{1}{2}$**	**$-\frac{\sqrt{3}}{2}$**	**IV**

Transform the wave y = A sin(B(x − C)) + D

Equation	Amplitude	Period	Phase shift	Midline
y = 3 sin x	**3**	**2π**	**none**	**y = 0**
y = sin 2x	**1**	**π**	**none**	**y = 0**
$y = 2\sin(\frac{1}{2}x) + 1$	**2**	**4π**	**none**	**y = 1**
$y = -\sin(x - \frac{\pi}{4})$	**1 (reflected)**	**2π**	**$\frac{\pi}{4}$ right**	**y = 0**

3 Write the rule fill in each blank from what you observed

1. On the unit circle, cos θ is the **x** -coordinate and sin θ is the **y** -coordinate.

2. sin θ is positive in Quadrants **I and II**. cos θ is positive in Quadrants **I and IV**.

3. For $y = A \sin(B(x - C)) + D$, the amplitude is **$|A|$** and the period is **$\frac{2\pi}{B}$**.

4. D moves the midline **up or down**. C shifts the wave **left or right**.

4 Check no tool: show your work

1. Find cos 120° exactly.

   $-\frac{1}{2}$

2. Find $\sin \frac{7\pi}{6}$ exactly.

   $-\frac{1}{2}$

3. Find the period of $y = 4 \sin 3x$.

   $\frac{2\pi}{3}$

4. Find the amplitude and midline of $y = -5 \sin x + 2$.

   5; y = 2

5 Reflect

Why does the sine graph repeat every 2π?

After a full turn of 2π radians the point is back where it started, so the heights repeat.